AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318

Applied Network Security

Final Examination
Spring Semester 2024/2025

Date: 12/June /2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 h

nstructions:

- Total Marks: 50
- Total Number of Questions:6
- Marks Distribution:
  - Question 1: 10
  - Question 2: 10
  - Question 3: 10
  - Question 4: 10
  - Question 5: 5
  - Question 6: 5
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

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Question 1: Answer the following questions [10 Marks]

1. **Which of the following is a characteristic of AES-GCM mode?**
   a) Provides only confidentiality
   b) Combines encryption and authentication
   c) Uses ECB chaining
   d) Requires RSA keys
2. **The Clipper Chip controversy was related to:**
   a) Key escrow in encryption
   b) Blockchain security
   c) Quantum-resistant algorithms
   d) Firewall configurations
3. **Which protocol uses port 443 by default?**
   a) HTTP
   b) FTP
   c) SSH
   d) HTTPS
4. **In the RSA algorithm, what ensures that *e* and *d* are inverses?**
   a) $ed \equiv 1 \bmod n$
   b) $ed \equiv 1 \bmod \phi(n)$
   c) *e* and *d* are prime
   d) *n* is a perfect square
5. **What is the primary role of a Registration Authority (RA) in PKI?**
   a) Issue certificates
   b) Verify user identities
   c) Revoke certificates
   d) Encrypt data
6. **Which hash function is resistant to length-extension attacks?**
   a) SHA-1
   b) SHA-256
   c) SHA-3
   d) MD5
7. **The Biba Model focuses on:**
   a) Confidentiality
   b) Integrity
   c) Availability
   d) Authentication
8. **Which TLS feature prevents downgrade attacks?**
   a) Session resumption
   b) Forward secrecy
   c) Cipher suite negotiation
   d) Extended Validation (EV) certificates
9. **What does SPF (Sender Policy Framework) protect against?**
   a) Phishing
   b) Email spoofing
   c) DDoS attacks
   d) SQL injection

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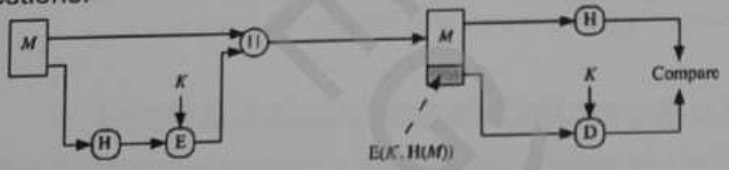
10. **In Diffie-Hellman, the security relies on the difficulty of:**
    a) Factoring large primes
    b) Solving discrete logarithms
    c) Breaking symmetric keys
    d) Reversing hash functions

estion 2: Answer the following questions [10 Marks]

a) List the steps Alice follows to generate her RSA public and private keys.(3 Marks)
b) If A sends an unencrypted message M to B, and also sends the Message Authentication Code, what two security services does the system provide? (2 marks)
c) Explain what happens if an attacker C intercepts the message from A to B, modifies M (e.g. changes M to N) and then forwards the modified message, with the original MAC attached, to B. (Include an explanation of whether B detects an attack) (2 marks)
d) Explain what happens if C changes both the message M (to N), and calculates and sends the new MAC based on N? (Include an explanation of whether B detects an attack) (3 marks)

estion 3: Answer the following questions [10 Marks]

a) List three security requirements for cryptographic hash functions and explain their importance. (2 Marks)
b) How does a digital signature provide message authentication and non-repudiation? Explain (2 Marks)
c) Explain the *birthday attack* on hash functions. How does it exploit the *birthday paradox*? (2 Marks)
d) Consider the security mechanism that shown in figure blow and answer the following questions:

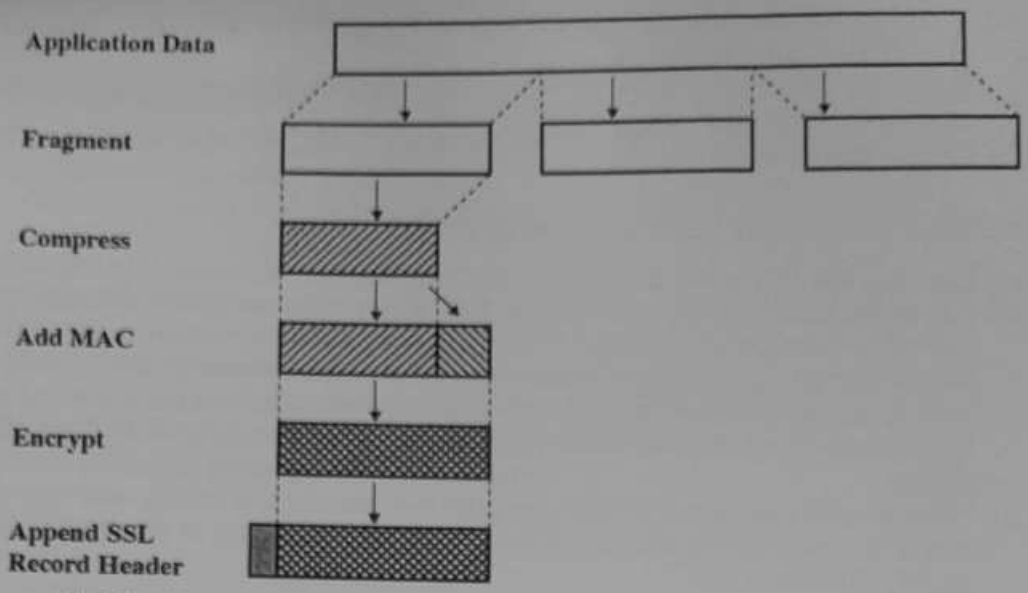


1) What is a security service that this mechanism provides? (2 Marks)
2) Define with details the weak collision resistant property (also called second pre- image resistant property) of a hash function.(2 marks)

stion 4: Answer the following questions [10 Marks]

a) What is the purpose of the X.509 standard? (1 Marks)
b) What is the major issue in end-to-end key distribution? How does the key hierarchy concept address that issue? (2 Marks)
c) The figure below shows the steps applied to application data by the SSL Record Protocol. (5 Marks)

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1) What two security services does the SSL Record Protocol provide? (1 Mark
2) List and explain an advantage and disadvantage of using SSL compared t using IPsec (2 Marks)
3) For normal (unsecure) web browsing, HTTP and TCP/IP are used. For secur web browsing, HTTPS is often used. Draw a protocol stack illustrating th protocols used when secure web browsing from application layer down t network layer, clearly showing the role of SSL. (2 marks)

d) Assume a digital signature is applied on the hash of a message (not the message itself), and A sends the message (MA) and digital signature to B. If the hash function is not weak collision resistant, then explain how an attacker C can forge A's digital signature on a new message from C (called MC). Make sure you explai why B cannot detect this forgery. (2 Marks)

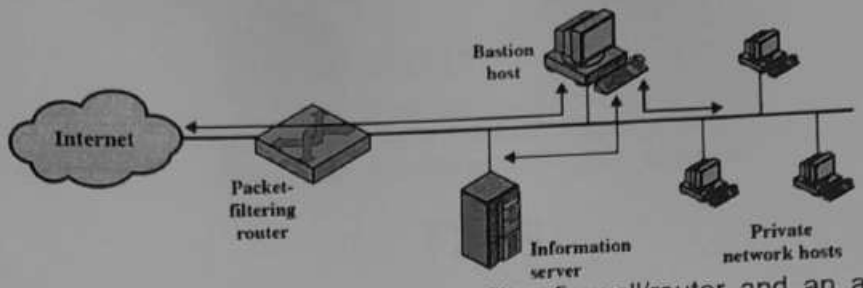
Question 5: Answer the following questions [5 Marks]

a) List two general approaches to dealing with replay attacks. (1 Marks)
b) How does salting improve password security? Provide an example. (1 marks)
c) How do rainbow tables speed up password cracking? How can salting mitigate this?(1 Marks)
d) Explain two weaknesses of password-based authentication. (1 marks)
e) Explain the Bell-LaPadula (BLP) model's two key properties and their purpose.1 Marks)

Question 6: Answer the following questions [5 Marks]

A single homed bastion host is one way to configure a network using both a packet filter firewall/router, as well as an application-level firewall (bastion host).

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a) Explain the difference between a packet filter firewall/router and an application level firewall. Include an advantage of each type of firewall. (2 marks)
b) Describe two rules that must be configured on the packet filter router in this configuration. (1 mark)
c) What is the difference between this configuration and a dual homed bastion host configuration? Explain how the dual homed configuration provides an advantage over the single homed configuration. (2 marks)

End of Questions

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